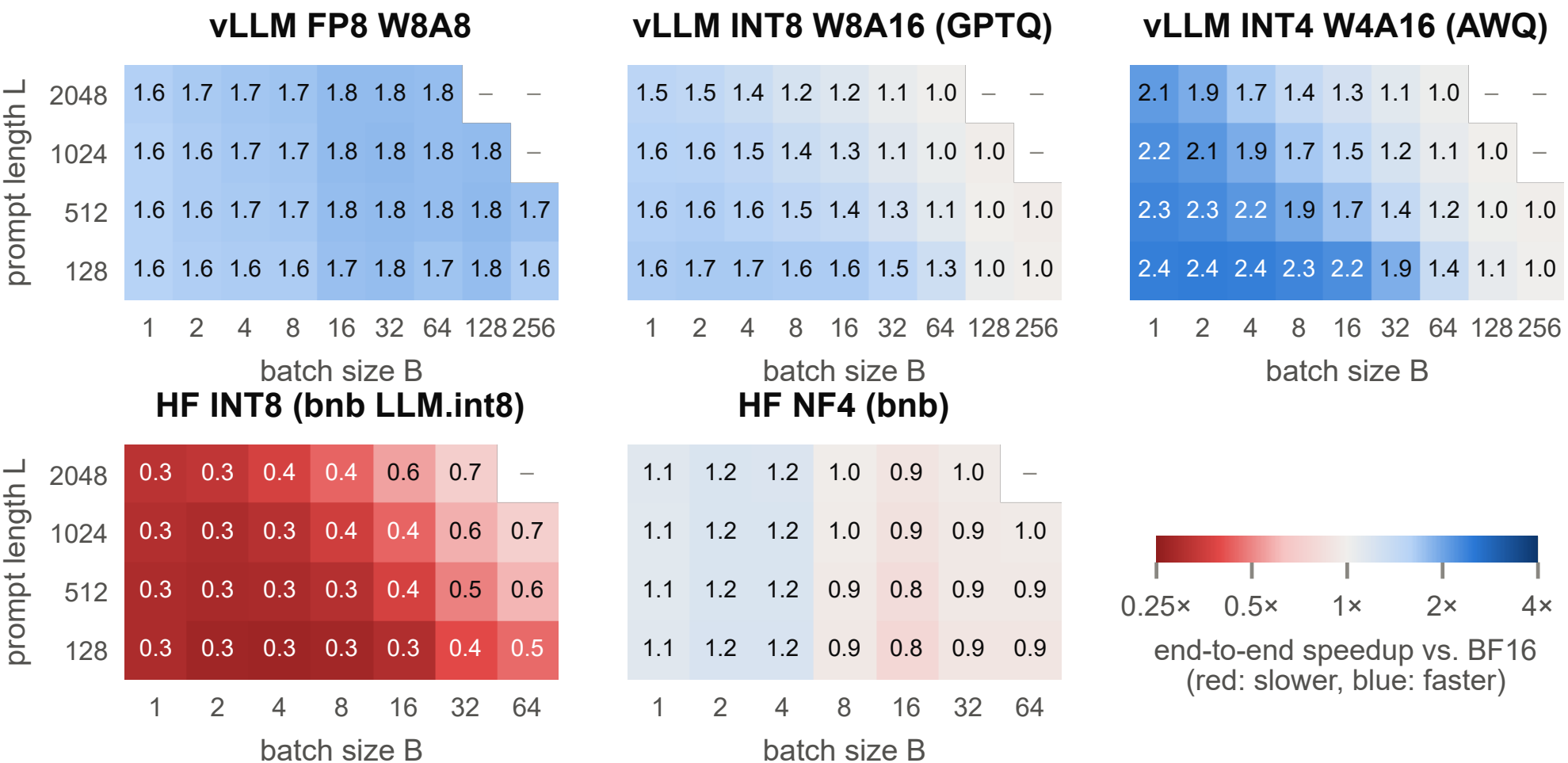
vLLM FP8 W8A8
vLLM INT8 W8A16 (GPTQ)
vLLM INT4 W4A16 (AWQ)
HF INT8 (bnb LLM.int8)
HF NF4 (bnb)
prompt length L
2048
1024
512
128
1 2 4 8 16 32 64 128 256
batch size B
1.6 1.7 1.7 1.7 1.8 1.8 1.8 – –
1.6 1.6 1.7 1.7 1.8 1.8 1.8 1.8 –
1.6 1.6 1.7 1.7 1.8 1.8 1.8 1.8 1.7
1.6 1.6 1.6 1.6 1.7 1.8 1.7 1.8 1.6
1.5 1.5 1.4 1.2 1.2 1.1 1.0 – –
1.6 1.6 1.5 1.4 1.3 1.1 1.0 1.0 –
1.6 1.6 1.6 1.5 1.4 1.3 1.1 1.0 1.0
1.6 1.7 1.7 1.6 1.6 1.5 1.3 1.0 1.0
2.1 1.9 1.7 1.4 1.3 1.1 1.0 – –
2.2 2.1 1.9 1.7 1.5 1.2 1.1 1.0 –
2.3 2.3 2.2 1.9 1.7 1.4 1.2 1.0 1.0
2.4 2.4 2.4 2.3 2.2 1.9 1.4 1.1 1.0
0.3 0.3 0.4 0.4 0.6 0.7 –
0.3 0.3 0.3 0.4 0.4 0.6 0.7
0.3 0.3 0.3 0.3 0.4 0.5 0.6
0.3 0.3 0.3 0.3 0.3 0.4 0.5
1 2 4 8 16 32 64
1.1 1.2 1.2 1.0 0.9 1.0 –
1.1 1.2 1.2 1.0 0.9 0.9 1.0
1.1 1.2 1.2 0.9 0.8 0.9 0.9
1.1 1.2 1.2 0.9 0.8 0.9 0.9
0.25× 0.5× 1× 2× 4×
end-to-end speedup vs. BF16
(red: slower, blue: faster)